

# TBA Delta Hedging

## Question

You went Long **\$100 million face value** of a 10-Year Fannie Mae 5.5% TBA contract at a market price of \$99.34375. Your trade settles on the upcoming front-month **Settlement Date of August 13, 2026**. Assume a constant prepayment speed of **6% CPR**.

1. How much value does a portfolio lose when interest rates go up by one basis point?
2. You want to create a duration-neutral macro hedge using a combination of 5-Year and 10-Year Pay-Fixed SOFR Swaps. Your risk mandate requires exactly 40% of the total hedge DV01 to come from the 5-Year Swap and 60% from the 10-Year Swap. Assume the following swap DV01s:

- \*\*5-Year Pay-Fixed SOFR Swap DV01:\*\* 0.0455% of notional per 1 bp
- \*\*10-Year Pay-Fixed SOFR Swap DV01:\*\* 0.0825% of notional per 1 bp

Using the TBA DV01 calculated in Part 1, Calculate the exact Notional Amounts of the 5- Year and 10-Year Swaps you must execute to create a perfect hedge.

| Input                   | Value     |                   |             |
|-------------------------|-----------|-------------------|-------------|
| Face Value              | \$100.00M |                   |             |
| TBA Price               | 99.34375  |                   |             |
| Annual Coupon Rate      | 5.50%     |                   |             |
| Remaining Term in Years | 10        |                   |             |
| Annual CPR              | 6%        |                   |             |
| Months Remaining        | 120       |                   |             |
| Initial Market Value    | \$99.34M  |                   |             |
| Monthly Coupon Rate     | 0.4583%   |                   |             |
| Monthly SMM             | 0.5143%   |                   |             |
| Monthly Payment         | \$1.0853M | <b>1 bps Jump</b> | <b>DV01</b> |
| Discount Factor         | 5.70%     | 5.71%             | \$0.0326M   |

|                                    |                  |                   |
|------------------------------------|------------------|-------------------|
| Face Value                         | \$100M           |                   |
| TBA DV01                           | \$0.0326M        |                   |
|                                    |                  |                   |
|                                    | <b>5Y Swap</b> ▼ | <b>10Y Swap</b> ▼ |
| Weight                             | 40%              | 60%               |
| Target Hedge DV01                  | \$0.0130M        | \$0.0195M         |
|                                    |                  |                   |
| Swap DV01 (% of Notional per 1 bp) | 0.0455%          | 0.0825%           |
| Hedge Notional                     | \$28.6417M       | \$23.6945M        |
| Hedge DV01 Generated               | \$0.0130M        | \$0.0195M         |
| Total Hedge DV01                   | \$0.0326M        |                   |
|                                    |                  |                   |
|                                    |                  |                   |
| Net DV01 After Hedge               | \$0.0000M        |                   |
| Solver Error                       | \$0.0000M        |                   |

## Experiment setup

|               | Naive Agent                       | Guided Agent                                           |
|---------------|-----------------------------------|--------------------------------------------------------|
| Corpus access | corpus                            | additional_corpus                                      |
| System prompt | <i>none</i>                       | Principal Fixed Income Quant Researcher + 5 guardrails |
| Tools         | list_files, read_file, run_python | list_files, read_file, run_python                      |

Each agent ran inside its own sandbox directory containing copies of only the corpora it was permitted to read.

# Answer — Naive Agent

## Step 1: TBA DV01 Calculation

**Assumption:** For a 10-year 5.5% coupon FNMA TBA prepaying at 6% CPR, the effective duration is approximately **4.50 years** (moderate extension/negative convexity reflecting a modest premium coupon with slow prepayments).

**Formula:**

$$DV01 = \text{Face Value} \times \frac{\text{Price}}{100} \times \text{Effective Duration} \times 0.0001$$

**Inputs:**

- Face Value = \$100,000,000
- Price = 99.34375 → 0.9934375
- Effective Duration = 4.50

**Calculation:**

$$DV01 = 100,000,000 \times 0.9934375 \times 4.50 \times 0.0001$$

$$DV01 = 99,343,750 \times 4.50 \times 0.0001 = 447,046,875 \times 0.0001$$

$$DV01_{TBA} \approx \$44,704.69 \text{ per 1bp}$$

Since you are **Long** the TBA, a **+1bp** rate increase causes a loss of approximately **\$44,704.69**.

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## Step 2: Hedge Allocation (40% / 60% Split)

You need a **Pay-Fixed** swap position (which gains value when rates rise) to offset the TBA's loss. Total hedge DV01 must equal the TBA DV01.

**Target DV01 contributions:**

- 5-Year Swap DV01 contribution (40%):

$$0.40 \times 44,704.69 = \$17,881.88$$

- 10-Year Swap DV01 contribution (60%):

$$0.60 \times 44,704.69 = \$26,822.81$$

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## Step 3: Solve for Notional Amounts

### 5-Year Swap:

$$\text{Notional}_{5Y} = \frac{17,881.88}{0.000455} = \$39,301,066$$

### 10-Year Swap:

$$\text{Notional}_{10Y} = \frac{26,822.81}{0.000825} = \$32,512,497$$

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## Summary

| Instrument         | DV01 Target                 | Notional (Pay-Fixed)  |
|--------------------|-----------------------------|-----------------------|
| TBA (Long)         | −\$44,704.69 (loss on +1bp) | \$100,000,000         |
| 5-Year Swap        | +\$17,881.88                | ≈ <b>\$39,301,066</b> |
| 10-Year Swap       | +\$26,822.81                | ≈ <b>\$32,512,497</b> |
| <b>Total Hedge</b> | <b>+\$44,704.69</b>         | —                     |

**Interpretation:** By entering into a **\$39.3 million** 5-Year Pay-Fixed swap and a **\$32.5 million** 10-Year Pay-Fixed swap, the combined DV01 of **+\$44,704.69** exactly offsets the TBA's **−\$44,704.69** DV01 loss from a 1bp rate increase, achieving a duration-neutral hedge with the mandated 40/60 risk allocation.

*(Note: Precise dollar figures depend on the effective duration assumption used for the TBA under the 6% CPR prepayment scenario — here assumed at 4.50 years, a typical value for a current-coupon-adjacent 5.5% 10-year MBS TBA.)*

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# Answer — Guided Agent

## Results

### Part 1 — TBA DV01 (Loss for +1bp rate move)

Using the deterministic cash-flow engine (10-yr FNMA 5.5%, 6% CPR, \$100mm face, price 99.34375):

- Market-implied annual discount rate solved: **5.7004%**
- Baseline PV (current): **\$99,343,750**
- PV after +1bp shock: **\$99,311,170.04**
- **TBA DV01  $\approx$  \$32,579.96 per 1bp**

Since you are **long** \$100mm face of the TBA, a 1bp rise in rates causes the position to **lose** approximately **\$32,580**.

### Part 2 — Duration-Neutral Swap Hedge (Pay-Fixed, since TBA is long/rates-down-favorable, you pay fixed to hedge rising rates)

Target hedge DV01 = \$32,579.96, split 40% / 60%:

- 5Y target DV01 =  $0.40 \times 32,579.96 = \$13,031.98 \rightarrow \text{Notional} = 13,031.98 / 0.000455$
- 10Y target DV01 =  $0.60 \times 32,579.96 = \$19,547.97 \rightarrow \text{Notional} = 19,547.97 / 0.000825$

### Final Hedge Notionals:

| Instrument                  | DV01 target | Notional                         |
|-----------------------------|-------------|----------------------------------|
| 5-Year Pay-Fixed SOFR Swap  | \$13,031.98 | $\approx$ <b>\$28,641,720.57</b> |
| 10-Year Pay-Fixed SOFR Swap | \$19,547.97 | $\approx$ <b>\$23,694,514.29</b> |

Check:  $(28,641,720.57 \times 0.000455) + (23,694,514.29 \times 0.000825) = \$13,031.98 + \$19,547.97 = \$32,579.96 = \text{TBA DV01} \checkmark$  ( $\Delta_{\text{net}} \approx 0$ , weights exactly 40%/60%).

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# Run metadata

| Field                 | Naive run                        | Guided run                       |
|-----------------------|----------------------------------|----------------------------------|
| Model                 | claude-sonnet-5                  | claude-sonnet-5                  |
| Effort                | low                              | low                              |
| First request ID      | req_011CdkPdM8d81futDvxaVZp1     | req_011CdkPefqYRB4F7mwHWh5rX     |
| Final response ID     | msg_011CdkPdQ4EpqcrVHXKhc83U     | msg_011CdkPg728vgKu1kXMmaQRF     |
| Start (UTC)           | 2026-08-05T22:28:45.957360+00:00 | 2026-08-05T22:29:04.002919+00:00 |
| End (UTC)             | 2026-08-05T22:29:03.996888+00:00 | 2026-08-05T22:29:30.168932+00:00 |
| Wall clock (s)        | 18.04                            | 26.167                           |
| API calls             | 1                                | 5                                |
| Tool calls            | 0                                | 4                                |
| Input tokens          | 1,048                            | 28,820                           |
| Output tokens         | 1,863                            | 2,473                            |
| Cache read tokens     | 0                                | 0                                |
| Cache creation tokens | 0                                | 0                                |
| Final stop reason     | end_turn                         | end_turn                         |

## API call trace — Naive Agent

| # | Request ID                   | Response ID                  | Stop     | In |
|---|------------------------------|------------------------------|----------|----|
| 1 | req_011CdkPdM8d81futDvxaVZp1 | msg_011CdkPdQ4EpqcrVHXKhc83U | end_turn | 10 |

# API call trace — Guided Agent

| # | Request ID                   | Response ID                  | Stop     | In |
|---|------------------------------|------------------------------|----------|----|
| 1 | req_011CdkPefqYRB4F7mwHWh5rX | msg_011CdkPegdQwdszDosNt2Xt2 | tool_use | 13 |
| 2 | req_011CdkPenFWKNauX7u3YBCvw | msg_011CdkPeo19hXGSuH3kcwERq | tool_use | 13 |
| 3 | req_011CdkPesT3Fcwom4F4eqywx | msg_011CdkPeuWKDevqa3HjVutrP | tool_use | 14 |
| 4 | req_011CdkPf1PnDKipRi8XRr3vh | msg_011CdkPf1oahLkrAidExfe5a | tool_use | 11 |
| 5 | req_011CdkPg6C381aBbbBaCEti8 | msg_011CdkPg728vgKu1kXMmaQRF | end_turn | 13 |

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